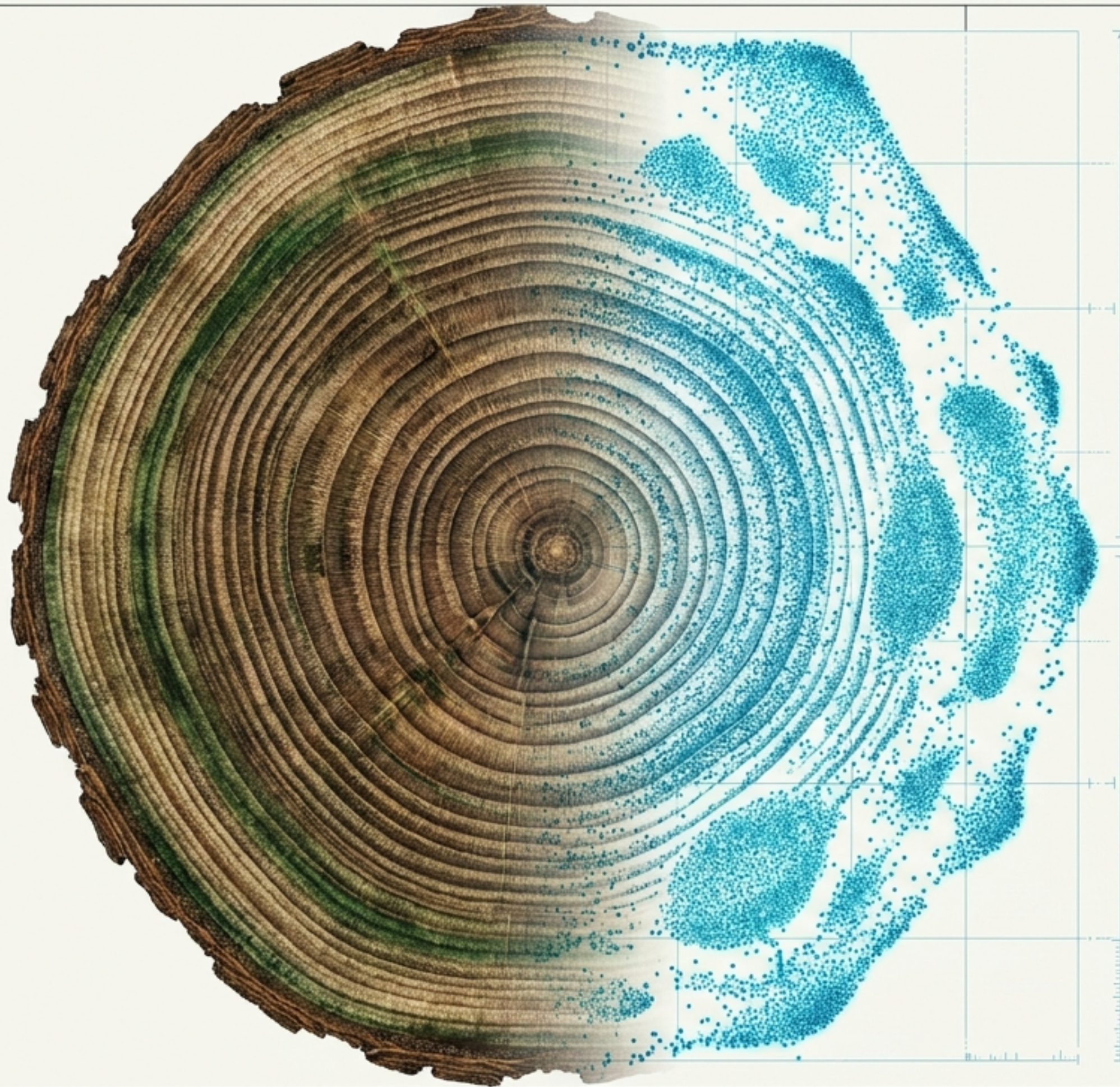
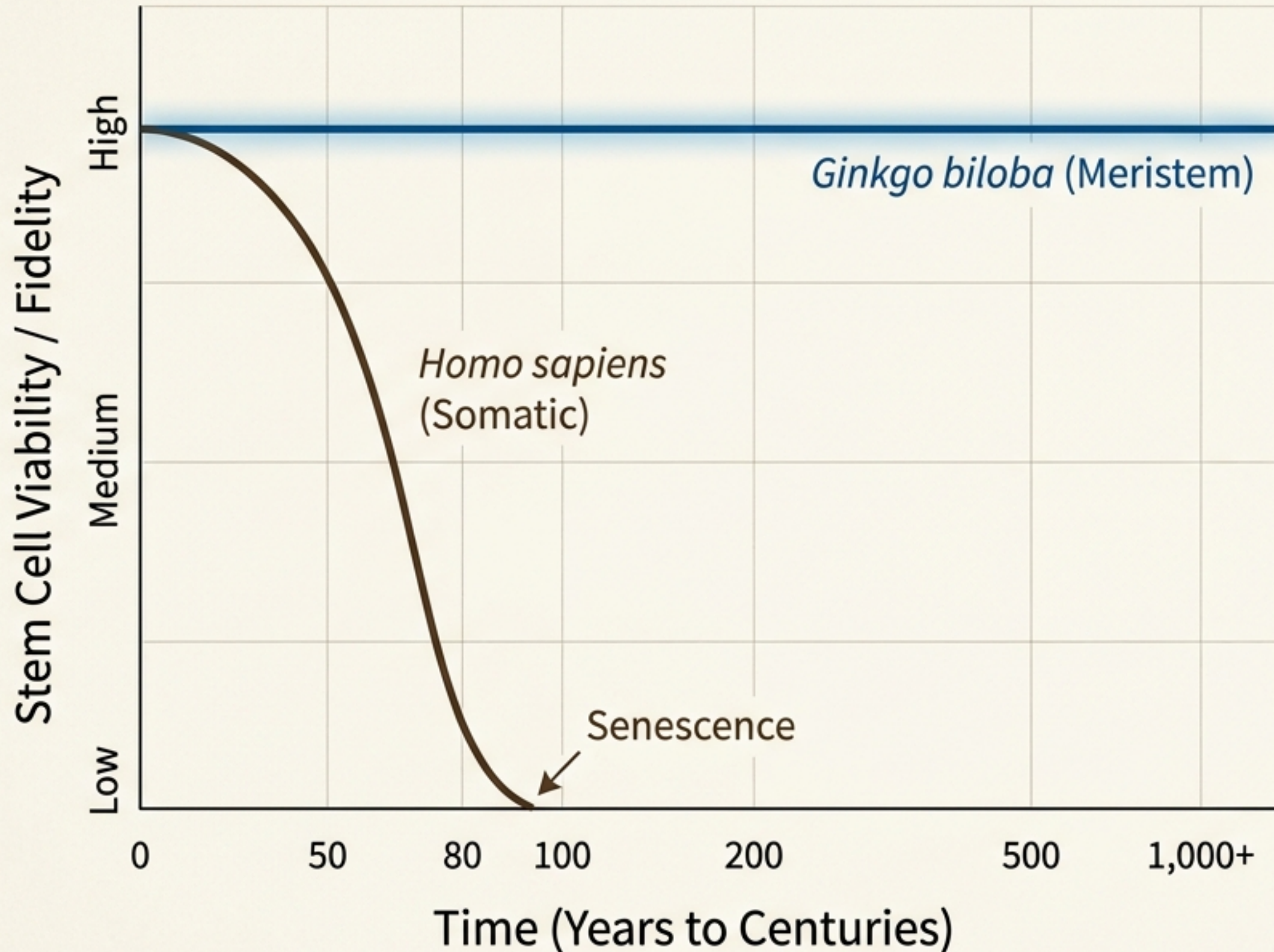




Decoupling Stem-Cell Maintenance from Cellular Senescence

A Comparative Investigation of Meristem Longevity and Quality Control in *Ginkgo biloba*

\$1.005M Grant Proposal Pitch



The Problem:

Human cellular maintenance machinery degrades over time, leading inevitably to stem-cell exhaustion, genomic decay, and senescence.

The Model:

Ginkgo biloba decouples time from aging, surviving millennia via flawless meristematic quality control and active cambial division.

The Proposal:

Decode this ancient biological blueprint using high-resolution single-nuclei transcriptomics to extract translatable mechanisms for human healthy aging.

The Engine of Immortality: Meristematic Homeostasis



The Self-Correcting Rheostat

The CLV-WUS spatial negative feedback loop dynamically balances stem cell renewal and differentiation. WUS maintains pluripotency non-autonomously, while secreted 13-aa CLV3 peptides prevent over-proliferation.

Stress Priming & Hypoxia

The meristem buffers against centuries of abiotic stress utilizing an endogenous hypoxic niche and the pre-accumulation of developmental priming transcription factors (e.g., STM, ATHB13, WRKY).

G-Protein Integration

Self-activating heterotrimeric G-proteins ($G\alpha/CT2$, $G\beta/AGB1$) couple with receptor-like kinases (FEA2) to calibrate meristem size without mammalian GPCR cascades.

Evolutionary Homology: The Baseline for Translation

Homo sapiens

Ginkgo biloba

Target Homology

Niche Maintenance

Mammalian Niche Decay

WUS/CLV Rheostat

Non-autonomous stem cell
signaling pathways

DNA Repair (NHEJ)

Fidelity drops,
mutations accumulate

Preserved fidelity via
Ku70/80 ring scaffold

XRCC6 / Ku70 architecture
(vWA, Ku-core, SAP domains)

Proteostasis

Autophagic decline,
toxic aggregates

Sustained
autophagic flux

ATG8 / MAP1LC3B ubiquitin-
like fold machinery

Stress Memory

Senescent arrest

Developmental priming,
hypoxic buffering

Stress-primed cell division
and metabolic adaptation

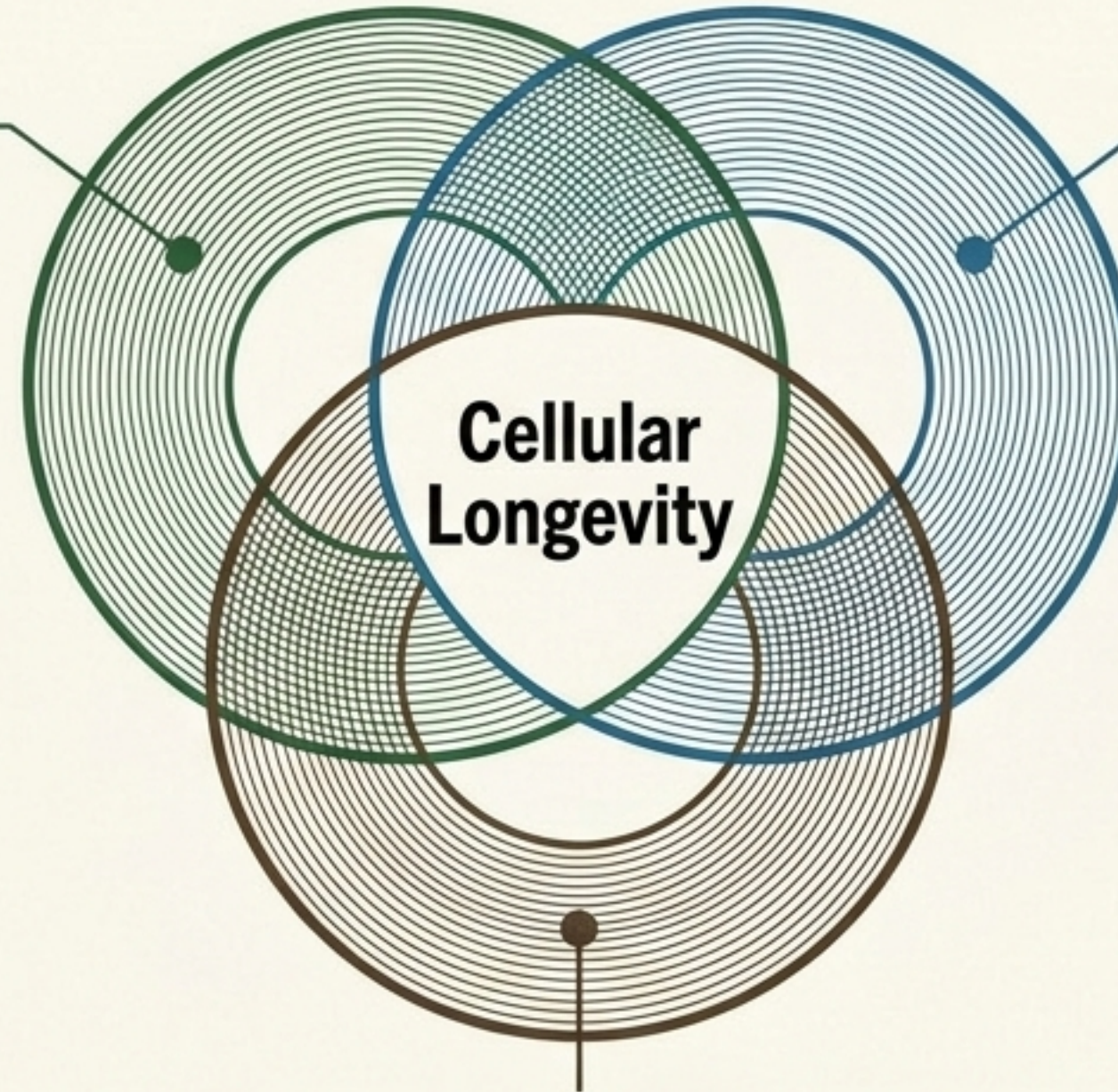
Integrated Research Aims

Aim 1: MAP (snRNA-seq)

Construct the first high-resolution transcriptomic atlas of the *Ginkgo biloba* Shoot Apical Meristem (SAM).

Aim 2: TIME (Heterochronic Axis)

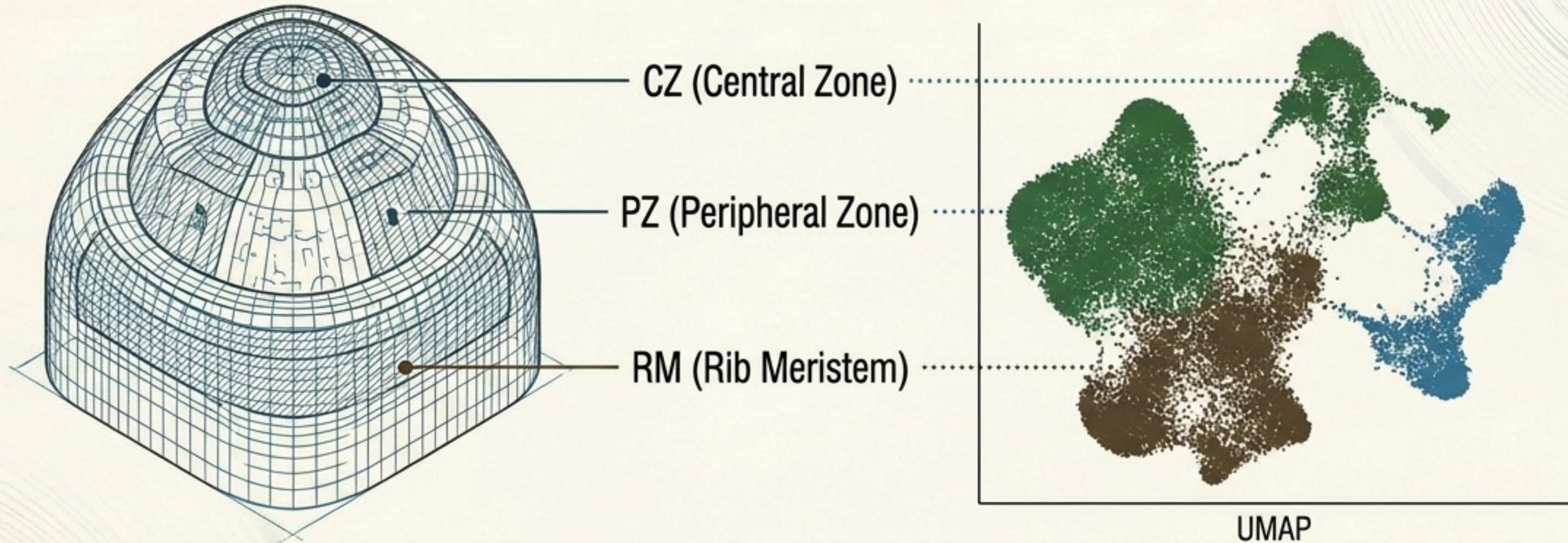
Define the age-invariant gene modules by comparing 1-year-old sapling meristems with 500-year-old ancient specimens.



Aim 3: FIX (Quality Control)

Characterize the exact functional preservation of DNA repair (Ku70/XRCC6) and proteostatic (ATG8) networks preventing mutational collapse.

Aim 1: Mapping the Ancient Meristem



Objective:

Execute the first-ever single-nuclei transcriptomic atlas of the Ginkgo SAM.

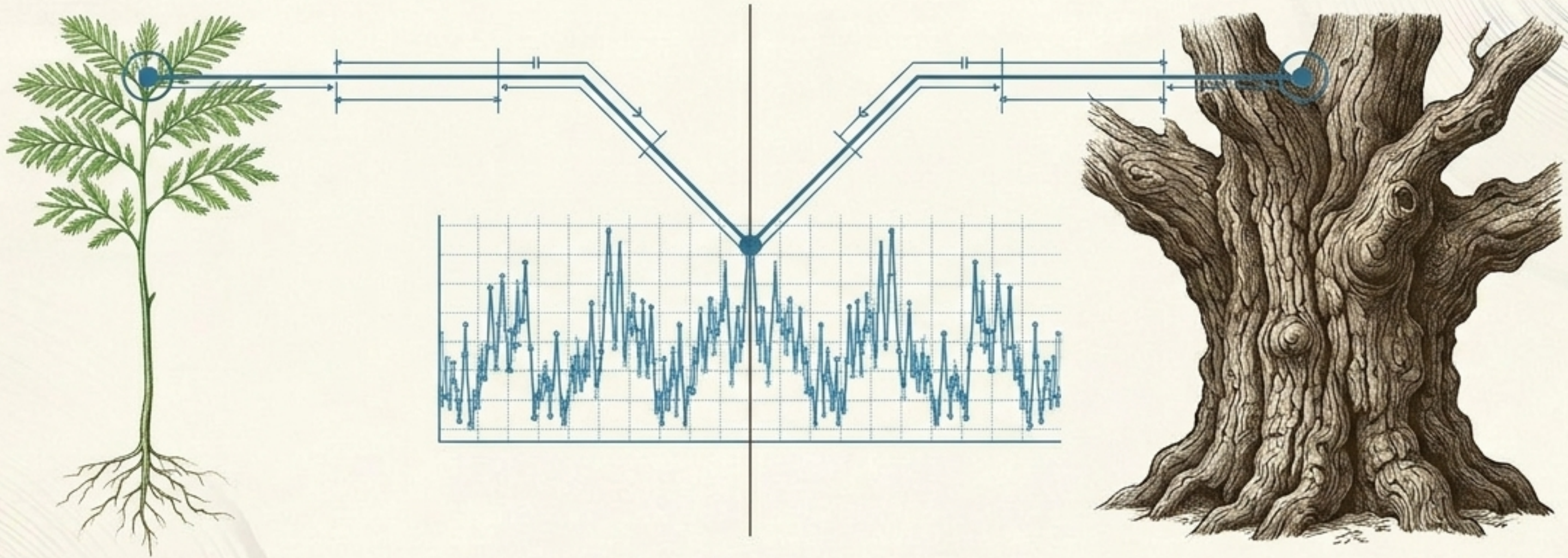
Key Deliverable:

Map the precise gene expression signatures maintaining the CLV-WUS spatial feedback loop and developmental plasticity across diverse cell types.

Focus:

Identify the overlapping domains of Cytokinin activation (AHK2/AHK4 receptors) and G-protein subunit signaling that dictate stem cell fate.

Aim 2: Decoding the Heterochronic Axis



Objective:

Answer the question: How does an ancient cell tell time?

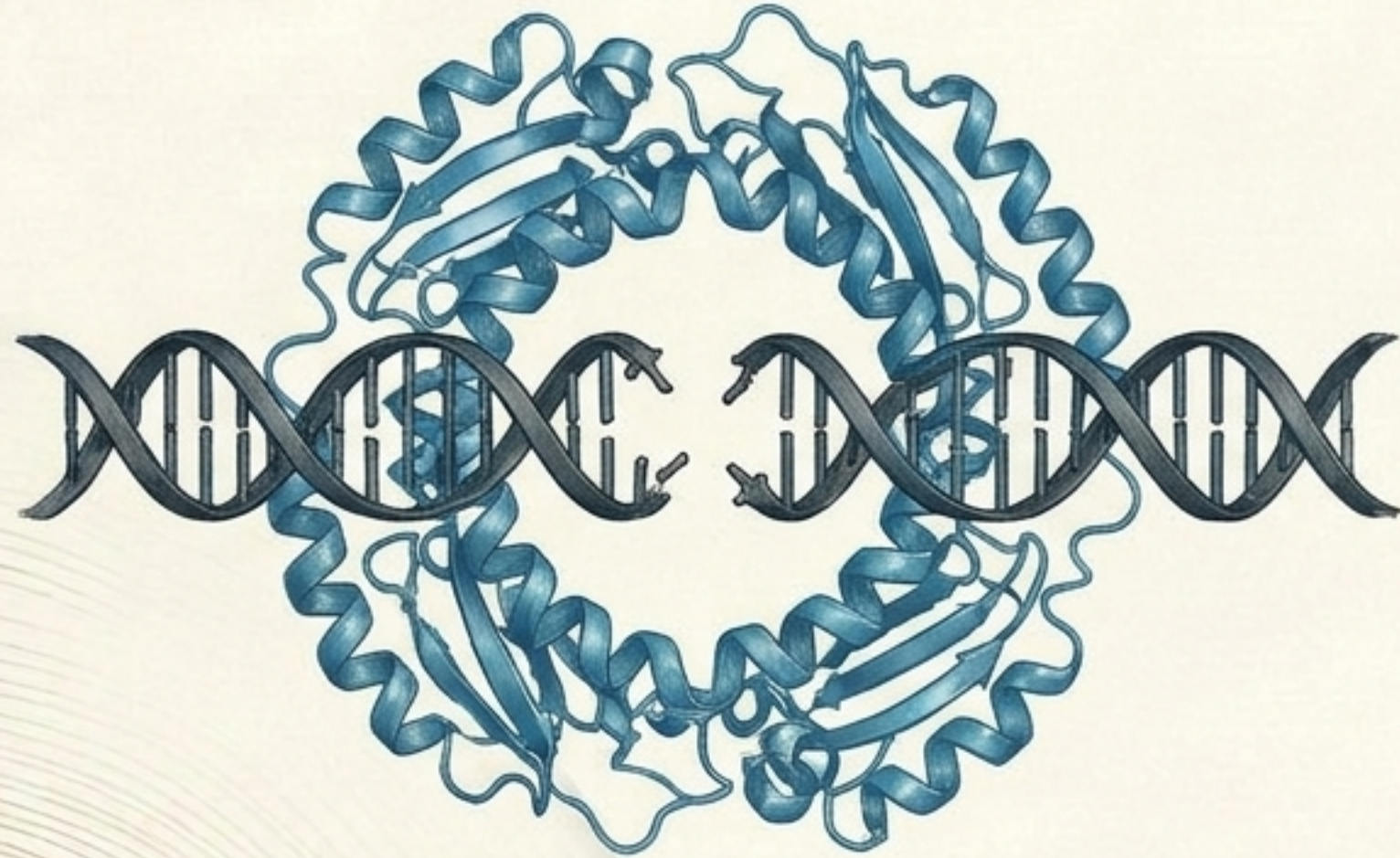
Method:

Sequence meristem and cambial nuclei from extreme age cohorts (Year 1 vs. Year 500+).

Hypothesis:

Long-lived trees enter a perpetual stress-primed state that halts biological aging clocks while maintaining perfect mitotic potential. We will isolate the specific gene modules responsible for this age-invariant transcriptomic signature.

Aim 3: Characterizing Quality Control Networks



Pilot Validation:

In silico bioinformatics confirm Ginkgo Ku70 shares 68.4% identity and 98% query coverage (E-value 0.0) with human XRCC6, perfectly preserving the vWA, Ku-core, and SAP domains across a billion years of evolution.



Objective:

Quantify how Ginkgo executes Non-Homologous End Joining (NHEJ) and autophagic flux to prevent somatic mutation accumulation and proteostatic collapse over hundreds of years.

Methodology: Overcoming the Secondary Metabolite Barrier



Woody Tissue



Cryo-milling



Tree-NEB Buffer Extraction



FANS Nuclei Sorting



10x Genomics Sequencing

The Challenge:

Extracting intact RNA from ancient, woody tissue is frequently compromised by rigid cell walls and abundant secondary metabolites.

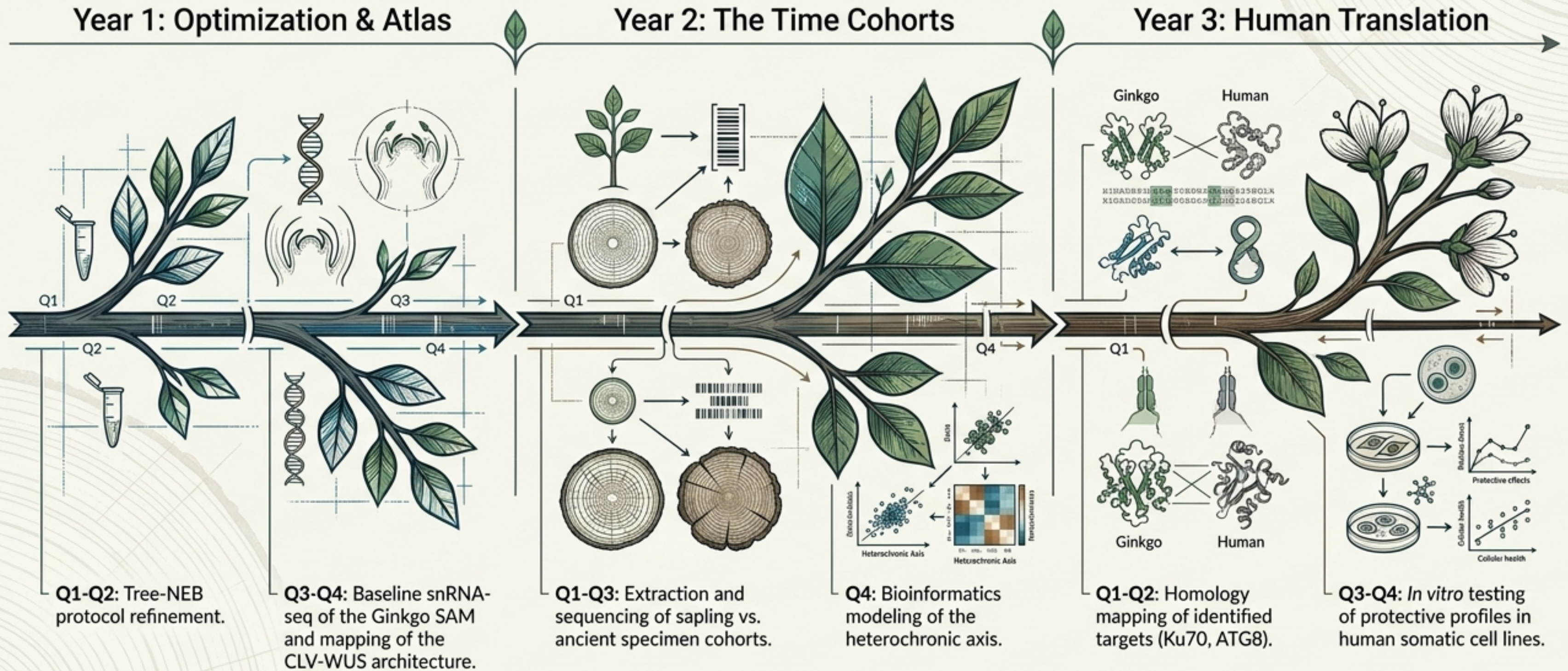
Our Solution:

The proprietary Tree-NEB single-nuclei protocol.

Impact:

Ensures high-yield, high-purity nuclei isolation compatible with FANS and 10x Genomics, derisking the entire transcriptomic pipeline.

3-Year Execution Timeline & Milestones



\$1.005M Itemized Budget Allocation

\$350k - Sequencing & Reagents

10x Genomics library prep, sequencing depth, Tree-NEB supplies.

\$155k - Equipment & FANS Time

Flow cytometry sorting and microscopy access.

\$300k - Personnel

Dedicated Post-doctoral researchers and Bioinformaticians.

\$100k - Human Cell Line Assays

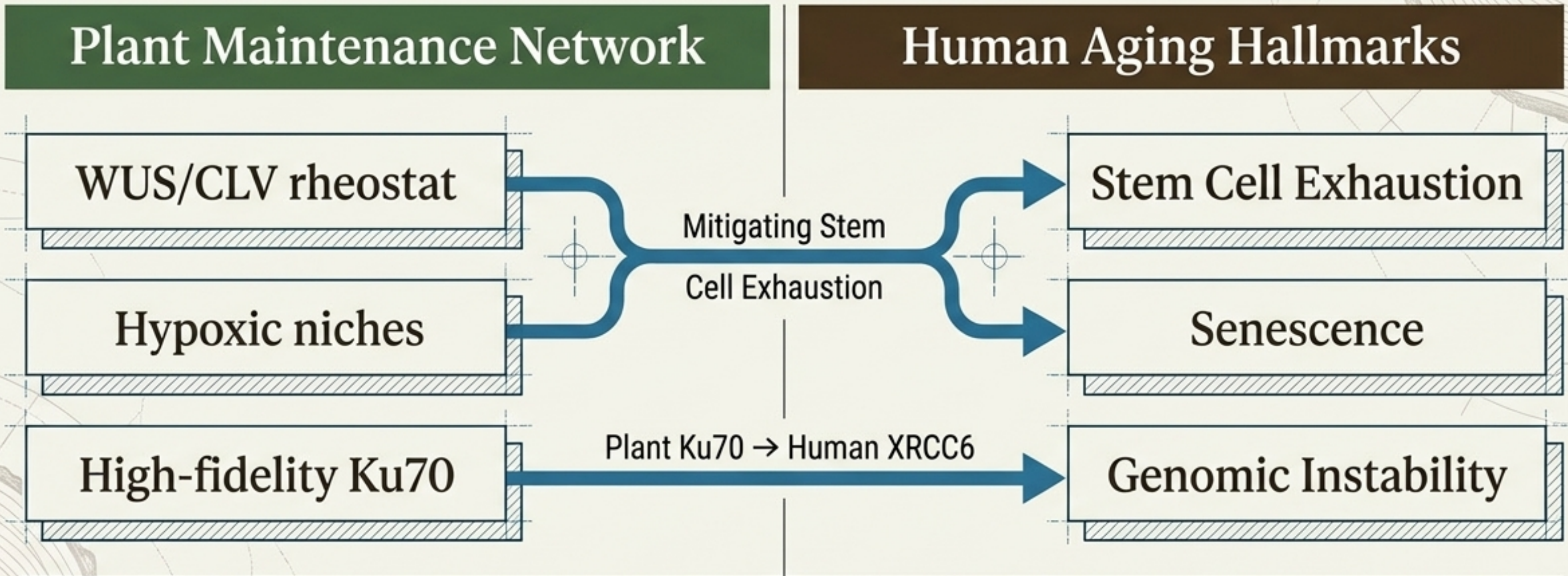
Phase 2 translational in vitro testing.

\$100k - Indirect Costs

Institutional overhead.

Summary Note: Lean, milestone-driven capital allocation maximizing raw data generation and direct translational output.

Synthesis: The Translational Bridge



The Paradigm Shift:

We are not merely studying botanical development; we are mining a biologically proven, naturally occurring database for human therapies. By decoding how the CLV-WUS loop and precise DNA repair maintain an immortal stem-cell niche, we isolate direct genetic and metabolic targets to combat human stem cell exhaustion and genomic instability.

Phase 2: Translational Human Cell Line Potential

The Next Frontier:

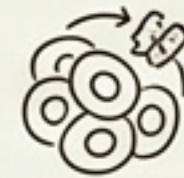
Validated mechanisms from Ginkgo will directly inform human in vitro models.

Key Intervention Targets:



Dermatological Resilience:

Leveraging plant stress-memory profiles to enhance skin barrier protection against UV and environmental stress.



Tissue Repair:

Preserving human progenitor niches through non-autonomous signaling (analogous to the WUS/CLV loop).



Stress Biosensors:

Developing dual-reporter live-cell assays to monitor proteostatic collapse and DNA repair efficacy.



From Ancient Roots to Human Longevity

The Vision: Decoupling cellular maintenance from the passage of time is not a theoretical impossibility—nature solved it millions of years ago.

The Impact: Funding this \$1.005M proposal establishes a bold new paradigm in aging research, moving beyond symptom management to unlock the ancient, fundamental blueprints of natural resilience.